9) Experiment to understand Logistic Regression for a given data set.

Program:

```
import numpy as np
import pandas as pd
df = pd.read_csv('/Social_Network_Ads.xlsx - Social_Network_Ads.csv')
df
df.head()
features = df.iloc[:, [2, 3]].values
label = df.iloc[:, 4].values
features
label
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
for i in range(1, 51):
    x_train, x_test, y_train, y_test = train_test_split(
        features, label, test_size=0.2, random_state=i
    )
    model = LogisticRegression()
    model.fit(x_train, y_train)
    test_score = model.score(x_test, y_test)
    train_score = model.score(x_train, y_train)
    print("Test", test_score, "Train", train_score, "Random State", i)
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=0
)
model = LogisticRegression()
model.fit(x_train, y_train)
model.score(x_train, y_train)
```

```
model.score(x_test, y_test)
from sklearn.metrics import classification_report
y_pred = model.predict(x_test)
print(classification_report(y_test, y_pred))
```

Output:

```
Test 0.8375 Train 0.84375 Random State 1
Test 0.8125 Train 0.853125 Random State 2
Test 0.8375 Train 0.85625 Random State 3
Test 0.9 Train 0.840625 Random State 4
Test 0.8625 Train 0.85 Random State 5
Test 0.8625 Train 0.859375 Random State 6
Test 0.8875 Train 0.8375 Random State 7
Test 0.8125 Train 0.853125 Random State 8
Test 0.8625 Train 0.8375 Random State 9
Test 0.9 Train 0.840625 Random State 10
Test 0.8125 Train 0.84375 Random State 11
Test 0.85 Train 0.853125 Random State 12
Test 0.8375 Train 0.8375 Random State 13
Test 0.8625 Train 0.85625 Random State 14
Test 0.85 Train 0.84375 Random State 15
Test 0.8625 Train 0.85625 Random State 16
Test 0.8375 Train 0.85625 Random State 17
Test 0.875 Train 0.834375 Random State 18
Test 0.85 Train 0.84375 Random State 19
Test 0.875 Train 0.84375 Random State 20
Test 0.8625 Train 0.834375 Random State 21
Test 0.875 Train 0.840625 Random State 22
Test 0.825 Train 0.84375 Random State 23
Test 0.875 Train 0.840625 Random State 24
Test 0.8 Train 0.846875 Random State 25
Test 0.85 Train 0.834375 Random State 26
Test 0.85 Train 0.840625 Random State 27
Test 0.8 Train 0.859375 Random State 28
Test 0.7875 Train 0.853125 Random State 29
Test 0.8625 Train 0.834375 Random State 30
Test 0.8625 Train 0.85625 Random State 31
Test 0.875 Train 0.853125 Random State 32
Test 0.8625 Train 0.84375 Random State 33
Test 0.8 Train 0.8625 Random State 34
Test 0.875 Train 0.83125 Random State 35
Test 0.8625 Train 0.853125 Random State 36
Test 0.8125 Train 0.865625 Random State 37
Test 0.8875 Train 0.840625 Random State 38
Test 0.875 Train 0.8375 Random State 39
Test 0.8375 Train 0.853125 Random State 40
Test 0.775 Train 0.859375 Random State 41
Test 0.8875 Train 0.8375 Random State 42
Test 0.775 Train 0.8625 Random State 43
Test 0.8375 Train 0.85625 Random State 44
Test 0.8375 Train 0.840625 Random State 45
Test 0.875 Train 0.846875 Random State 46
Test 0.9125 Train 0.83125 Random State 47
Test 0.825 Train 0.84375 Random State 48
Test 0.7625 Train 0.8625 Random State 49
```

	precision	recall	f1-score	support
0	0.92	0.97	0.94	58
1	0.89	0.77	0.83	22
accuracy			0.91	80
macro avg	0.91	0.87	0.89	80
weighted avg	0.91	0.91	0.91	80

Result:

Experiment is conducted to understand Logistic Resgression for the given data set.